Math 110 — Week 8

Solutions

HW8 P1.

(a) **Idea.** Differentiation lowers degree, so enough derivatives kill any polynomial; the index is the first power that kills x^n, the slowest vector to die.

D lowers degree by 1 and kills constants, so $D^{n+1}p = 0$ for every $p \in \mathcal{P}_n(\mathbb{C})$, while $D^n x^n = n! \neq 0$. So D is nilpotent of index $n+1$.

(b) $E(0, D) = \operatorname{null} D =$ constants, so the geometric multiplicity is 1. $G(0, D) = \mathcal{P}_n(\mathbb{C})$ since D is nilpotent, so the algebraic multiplicity is $n+1$.

(c) **Idea.** One Jordan block means one chain. Start from the vector that takes longest to die (x^n) and differentiate repeatedly; dividing by factorials makes each step land on the next basis vector with coefficient 1.

$f_k = x^k/k!$ for $k = 0, \ldots, n$: $Df_0 = 0$ and $Df_k = x^{k-1}/(k-1)! = f_{k-1}$. In the basis $f_0, f_1, \ldots, f_n$, column k of the matrix has a single 1 in row $k-1$: the matrix is the single block $J_{n+1}(0)$.

HW8 P2.

(a) **Idea.** Translate the data into blocks: geometric multiplicity = number of blocks, algebraic = total size, largest block = exponent in m.

Number of blocks = geometric multiplicity, total size = algebraic multiplicity: $J_3(2) \oplus J_1(-1) \oplus J_1(-1)$,

$$\begin{pmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix}, \qquad m = (x-2)^3(x+1).$$

(b) Two blocks for 2 with total size 3: $J_2(2) \oplus J_1(2) \oplus J_1(-1) \oplus J_1(-1)$, i.e. the matrix in (a) with the $(2,3)$ entry changed to 0. The largest block for 2 has size 2, so $m = (x-2)^2(x+1)$.

HW8 P3.

(a) $N = J - I$ has $Ne_1 = 0$, $Ne_2 = e_1$, $Ne_3 = e_2$. So $\operatorname{null} N = \operatorname{span}(e_1)$; $N^2 e_3 = e_1$ and $N^2 e_1 = N^2 e_2 = 0$, so $\operatorname{null} N^2 = \operatorname{span}(e_1, e_2)$; $N^3 = 0$, so $\operatorname{null} N^3 = \mathbb{F}^3$.

(b) $G(1, J) = \operatorname{null}(J - I)^3 = \mathbb{F}^3$. Geometric multiplicity $\dim \operatorname{null}(J - I) = 1$; algebraic multiplicity $\dim G(1, J) = 3$.

(c) **Idea.** Split J into a scalar part and a nilpotent part that commute, then use the binomial theorem; it stops at N^2 because $N^3 = 0$.

$J = I + N$ with $IN = NI$ and $N^3 = 0$, so

$$J^m = I + mN + \binom{m}{2} N^2 = \begin{pmatrix} 1 & m & \binom{m}{2} \\ 0 & 1 & m \\ 0 & 0 & 1 \end{pmatrix}$$

(with $\binom{m}{2} = \frac{m(m-1)}{2}$; for $m = 1$ this is J).

HW8 P4.

(a) **Idea.** Null spaces of a block diagonal matrix add block by block, and for one block $J_j(0)^k$ is a shift by k places.

For one block $J_j(0)$, $\dim\operatorname{null} J_j(0)^k = \min(k, j)$. Add over the blocks of sizes 3 and 2: $k = 0$: 0; $k = 1$: $1+1 = 2$; $k = 2$: $2 + 2 = 4$; $k = 3$: $3 + 2 = 5$; $k = 4$: $3 + 2 = 5$.

(b) **Idea.** Differences of the d_k count blocks of size $\geq k$; the total dimension pins down the largest size.

The number of blocks of size $\geq k$ is $\dim\operatorname{null} N^k - \dim\operatorname{null} N^{k-1}$: $2 - 0 = 2$ blocks of size ≥ 1, $4 - 2 = 2$ of size ≥ 2, $5 - 4 = 1$ of size ≥ 3. So there are two blocks, both of size at least 2, exactly one of size at least 3. The three numbers alone do not fix the larger size ($J_4(0) \oplus J_2(0)$ on $\mathbb{C}^6$ also gives $2, 4, 5$), so we use that N acts on the same 5-dimensional space as in (a): then $\operatorname{null} N^3 = V$ and $N^3 = 0$, so no block has size above 3. Hence the sizes are 3 and 2. (Equivalently: the sizes are 2 and $s \geq 3$, and $2 + s = \dim V = 5$.)

HW8 P5.

Idea. For each eigenvalue, make the table of d_k and take differences (Prop. 77).

Eigenvalue 1: $d_1, d_2, d_3 = 1, 2, 2$, so 1 block of size ≥ 1, 1 of size ≥ 2, none of size ≥ 3: $J_2(1)$. Eigenvalue 2: $d_1, \ldots, d_4 = 2, 3, 4, 4$, so 2 blocks of size ≥ 1, 1 of size ≥ 2, 1 of size ≥ 3, none of size ≥ 4: $J_3(2)$ and $J_1(2)$. Check: $2 + 3 + 1 = 6$. So $T \sim J_2(1) \oplus J_3(2) \oplus J_1(2)$.

HW8 P6.

(a) $\chi = (x-4)^5(x+1)$. (b) $m = (x-4)^3(x+1)$ (the largest block for 4 has size 3). (c) Eigenvalue 4: geometric 2 (two blocks), algebraic 5; eigenvalue -1: geometric 1, algebraic 1. (d) Not diagonalizable: m has the repeated root 4 (equivalently, $2 \neq 5$).

HW8 P7.

Idea. Everything about a block diagonal matrix splits into blocks, so count block by block; for one block the null space of $J_j(0)^k$ is easy because $J_j(0)^k$ is a shift.

Choose a basis in which N is in Jordan form with blocks $J_{j_1}(0), \ldots, J_{j_r}(0)$. N^k is block diagonal with blocks $J_{j_i}(0)^k$, so $\dim\operatorname{null} N^k = \sum_i \dim\operatorname{null} J_{j_i}(0)^k$. On one block with basis $e_1, \ldots, e_j$, $J_j(0)^k e_l = e_{l-k}$ for $l > k$ and 0 for $l \leq k$; so $\operatorname{null} J_j(0)^k = \operatorname{span}(e_1, \ldots, e_{\min(k,j)})$ and $\dim\operatorname{null} J_j(0)^k = \min(k, j)$. Then $\min(k, j) - \min(k-1, j)$ is 1 if $j \geq k$ and 0 if $j < k$. Summing over the blocks, $\dim\operatorname{null} N^k - \dim\operatorname{null} N^{k-1}$ counts the blocks with $j_i \geq k$.

G1.

Idea. Get the eigenvalues from a characteristic polynomial, then make the d_k table, then build chains from the top.

The upper-left block has trace 4 and determinant 4, so $\chi_A = (x-2)^2 \cdot (x-2) = (x-2)^3$: only eigenvalue 2.

$$A - 2I = \begin{pmatrix} -1 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

has rank 1, so

$$E(2, A) = \{y = x\} = \operatorname{span}\big((1, 1, 0), (0, 0, 1)\big)$$

and $d_1 = 2$. $(A-2I)^2 = 0$, so $d_2 = 3$ and $G(2, A) = \mathbb{C}^3$. Blocks: 2 of size ≥ 1, 1 of size ≥ 2: $J_2(2) \oplus J_1(2)$, $m_A = (x-2)^2$. Jordan basis: top $e_1 \notin \operatorname{null}(A - 2I)$, bottom $(A - 2I)e_1 = (-1, -1, 0)$; then the eigenvector e_3, independent of that bottom. In the basis $(-1, -1, 0), e_1, e_3$ the matrix is $J_2(2) \oplus J_1(2)$ (check: $Ae_1 = (1, -1, 0) = 2e_1 + (-1, -1, 0)$).

G2.

Idea. Same as for eigenspaces: S commutes with every polynomial in T, so it preserves null spaces of polynomials in T.

$G(\lambda, T) = \operatorname{null}(T - \lambda I)^n$. If $(T - \lambda I)^n v = 0$, then $(T - \lambda I)^n Sv = S(T - \lambda I)^n v = 0$.

G3.

(a) **Idea.** $(\lambda I + N)v = 0$ says v is an eigenvector of N for $-\lambda$, and 0 is the only eigenvalue of N.

If $\lambda \neq 0$ and $(\lambda I + N)v = 0$, then $Nv = -\lambda v$; since $-\lambda$ is not an eigenvalue of N, $v = 0$. So $\lambda I + N$ is injective, hence invertible. If $\lambda = 0$, N is not injective ($V \neq 0$ and $N^k = 0$). Explicitly, for $\lambda \neq 0$ the inverse is

$$(\lambda I + N)^{-1} = \lambda^{-1} \sum_{j=0}^{n-1} (-N/\lambda)^j.$$

(b) **Idea.** With one eigenvalue there is one generalized eigenspace, and it is everything.

By Prop. 73, $V = G(\lambda, T) = \operatorname{null}(T - \lambda I)^n$, so $(T - \lambda I)^n = 0$.

G4.

Idea. Block sizes add to the algebraic multiplicity 4; the largest is the exponent 2 in m_T.

Sizes with largest 2 and sum 4: $\{2, 2\}$ or $\{2, 1, 1\}$. So $J_2(2) \oplus J_2(2)$ or $J_2(2) \oplus J_1(2) \oplus J_1(2)$. They have the same χ_T and m_T; $\dim E(2, T)$ (2 versus 3) tells them apart.

G5.

Idea. Over $\mathbb{C}$, diagonalizable means $E(\lambda, T) = G(\lambda, T)$ for every λ.

If $0 \neq v \in G(\lambda, T)$, the hypothesis gives $Tv = \mu v$ for some μ; then $0 = (T - \lambda I)^n v = (\mu - \lambda)^n v$, so $\mu = \lambda$. Hence $G(\lambda, T) \subseteq E(\lambda, T)$, so $G(\lambda, T) = E(\lambda, T)$ for each eigenvalue. Then

$$V = \bigoplus G(\lambda_i, T) = \bigoplus E(\lambda_i, T)$$

(Prop. 73), and a union of bases of the eigenspaces is an eigenbasis.